

TFPT verification suite: the journey of ~300 machine-checked scripts

v1-v23 - Foundations

carrier $D5+A3+\mu_4$, E8 glue, α^{-1} fixed point,
anchor $a=(1,1,2) \rightarrow \{c_3, g_{\text{car}}\}$ reduce to $\{a, \pi\}$

v24-v53 - Standard-Model readouts

ϕ_0 mass ladder, flavor (Q,K,R,L), lepton/quark
ratios, hypercharge, compiler core (5,3)

v54-v100 - Seam = horizon

one-sided Gauss-Bonnet $c_3=1/(8\pi)$, Coxeter-30 cycle,
gapped attractor $(2/3)^6$, frozen registry + null-MC

v101-v140 - Horizon + flavor geometry

Nariai=anchor, monodromy= $W(A_3)=S_4$, cusp
weights $\{0, 1/3, 2/3\}$, H^1 cohomology, canonical map

v141-v158 - R1-R5 + premise (A)

deck selection, EH mechanism, No-Unit Theorem,
simple-current (E8)₁, free $c=8$ fixed point isolated/stable

v159-v169 - PyR@TE cross-checks

SM gauge/Yukawa/Higgs RGEs confirmed ($b_1=41/10$),
 Λ_{QCD} , m_H near-criticality, η_B leptogenesis

v170-v174 - AQFT bridges

E8 slice compression, OS moment/Sugawara, trace
anomaly seed $4/3$, strong-CP Pfaffian, Fock readings

v175-v181 - AQFT closure -> geometric bedrock

net existence + full-cone RP [E];
residual -> one premise: carrier μ_4 clock = seam conformal deck

v182-v213 - F_transfer functor + frontier

reviewer residual map; F_transfer = one typed
functor (Koide, η_B , axion, m_p/m_e) + machine guard

v214-v218 - Pillowcase + Sheet Diamond

QGeo pillowcase (cross-ratio 2 $\Rightarrow j=1728$), four
marks from Gauss-Bonnet, the Sheet Diamond axis geometry

v219-v237 - Icosahedral capstone

McKay why $\{2,3,5\}$, CM norms $41/7$, CP triality, Kleinian
seam, $\det K=1$ = Kitaev E8; the (2,3,5) Brieskorn generates ALL

v238-v261 - NCG / Modular Spectral Closure

96-dim spectral triple, Dirac = covariance induction,
cutoff = KMS weight, seam/carrier/E8 on one K3: one relative object

v262-v275 - Frontier closure + QFT4D fork

F_{QCD} m_p/m_e , M_ν seesaw, S_{pert} (EG, 1-loop + gauge
betas), scale over-determination, the QG.AMB roadmap

v276-v299 - The Gral: SEAM.EQUIV.01 + Flat-Away

seam = (E8)₁ at $\tau=i$ named as ONE theorem;
both routes reduced to one shared input (Flat-Away), heat a_2 proved+Lean (v295/v296)

v300-v302 - Closing arc: residual pinned

Flat-Away hard+pin from (E8)₁ (v300); Route A invertible
via free fermions (v301); gap = derived $6\ln(3/2)>0$ (v302); no TFPT-internal assumption left